15. Find the determinant of $M=\begin{pmatrix}3&0&0\\0&3&0\\0&0&3\end{pmatrix}$.

A. -1
B. 3
C. 9
D. 27

16. Given that $A=\begin{bmatrix}1&1\\2&2\end{bmatrix}$ and $B=\begin{bmatrix}-1&1\\1&-1\end{bmatrix}$, find AB.

A. $\begin{bmatrix}1&0\\0&1\end{bmatrix}$ B. $\begin{bmatrix}0&1\\1&0\end{bmatrix}$ C. $\begin{bmatrix}0&0\\0&0\end{bmatrix}$ D. $\begin{bmatrix}1&1\\1&1\end{bmatrix}$

17. Find the inverse of the matrix $A=\begin{pmatrix}2&3\\4&5\end{pmatrix}$.

A. $-2\begin{pmatrix}5&-3\\-4&2\end{pmatrix}$ B. $-\frac{1}{2}\begin{pmatrix}5&-3\\-4&2\end{pmatrix}$

C. $2\begin{pmatrix}5&-3\\-4&2\end{pmatrix}$ D. $\frac{1}{2}\begin{pmatrix}5&-3\\-4&2\end{pmatrix}$

18. Consider the systems, each consisting of m linear equations in n variables. Which one of the following is correct?

I. If $m<n$, then all such systems have a solution.
II. If $m>n$, then none of these systems has a solution.
III. If $m=n$, then there exists a system which has a solution.

A. I, II and III are true.
B. Only II and III are true.
C. Only III is true.
D. None of them is true.

19. Find the solution to the system of equations $\begin{bmatrix}2&5\\-4&3\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}2\\-30\end{bmatrix}$.

A. $x=6, y=2$
B. $x=-6, y=2$
C. $x=-6, y=-2$
D. $x=6, y=-2$

20. For what value of p will the following set of equations have no solution?
$2x+3y=5$
$Bx+py+10$

A. 4.5
B. 3.5
C. 2.5
D. 1.0

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SEMESTER I QUIZ I, 2020/2021
4 YR BACHELOR OF MATHEMATICS AND SCIENCE EDUCATION
4 YR BME / BSE YR 1
MAT 101: ALGEBRA AND TRIGONOMETRY
EXAM CODE: 01

NAME: .....

REGISTRATION NO.:..... SIGNATURE:.....

STUDY CENTRE:.....TIME ALLOWED: 40 MINUTES

INSTRUCTION: *Answer all questions by circling the correct option in ink.*

1. Which of the following sets are infinite?

I. {factors of 100}
II. {multiples of 3}
III. {prime numbers}

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

2. If $P=\{x:-5<x<5, x\in\text{Integer}\}$, $Q=\{x:-3\le x<6, x\in\text{Integer}\}$ and $R=\{x:-4<x\le 5, x\in\text{Integer}\}$, find $(P\cap Q)\cup(P\cap R)$.

A. $\{x:-4<x<5, x\in\text{Integer}\}$
B. $\{x:-4\le x<5, x\in\text{Integer}\}$
C. $\{x:-4<x\le 5, x\in\text{Integer}\}$
D. $\{x:-4\le x\le 5, x\in\text{Integer}\}$

3. Which of the following represents associativity of union of sets?

A. $P\cup(Q\cup R)$
B. $(P\cap Q)\cap R$
C. $P\cup(Q\cap R)$
D. $(P\cap Q)\cup R$

4. If A, B and C are any subsets of a universal set U. Then, $(A\cup B\cup C)'=A\cap B'\cap C'$.

A. True
B. False

5. Given the sets $A=\{2,3,4\}$ and $B=\{4,5\}$, which of the following is true?
   I. $A\times B=\{(2,4),(2,5,),(3,4),(3,5),(4,4),(4,5)\}$
   II. $B\times A=\{(4,2),(4,3),(4,4),(5,2),(5,3),(5,4)\}$
   III. $B\times B=\{(4,4),(4,5),(5,4),(5,5)\}$
   A. I and II only
   B. I and III only
   C. II and III only
   D. I, II and III

6. If the set $A=\{1,2,3,4,5\}$ and $B=\{w,x,y,z\}$, how many elements are there in the power set of $A\times B$?
   A. 0
   B. 2
   C. 20
   D. 1,048,576

7. The set $S=\{0,1,2,3,4,\cdots\}$ is called the set of ——— numbers.
   A. natural
   B. counting
   C. whole
   D. real

8. Which of the following equations have no solution in the set of integers?
   I. $5(x+2)=7$
   II. $12x+3=-9$
   III. $3(x+2)=4(x+2)$
   A. I only
   B. II only
   C. I and III only
   D. I, II and III

9. Express the decimal $0.12\dot{3}\dot{1}$ in the form $\frac{a}{b}$, where a and b are integers.
   A. $\frac{1.299}{9.9}$
   B. $\frac{12.99}{99}$
   C. $\frac{129.9}{990}$
   D. $\frac{1299}{9900}$

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10. Rationalize: $\frac{7-\sqrt{7}}{5+\sqrt{5}}$.
    A. $\frac{35+7\sqrt{5}+5\sqrt{7}+\sqrt{35}}{20}$
    B. $\frac{35-7\sqrt{5}-5\sqrt{7}+\sqrt{35}}{20}$
    C. $\frac{35+7\sqrt{5}-5\sqrt{7}-\sqrt{35}}{20}$
    D. $\frac{-35+7\sqrt{5}+5\sqrt{7}-\sqrt{35}}{20}$

11. Simplify: $2+5\times 8-3\div 6\times 2$.
    A. 18
    B. 28
    C. 35
    D. 41

12. Find $A+B$, if $A=\begin{bmatrix}2 & 5\\ 7 & 0\end{bmatrix}$ and $B=\begin{bmatrix}6 & 5\\ 1 & 1\end{bmatrix}$.
    A. $A+B=\begin{bmatrix}8 & 8\\ 10 & 1\end{bmatrix}$
    B. $A+B=\begin{bmatrix}10 & 30\\ 0 & 1\end{bmatrix}$
    C. $A+B=\begin{bmatrix}8 & 10\\ 8 & 1\end{bmatrix}$
    D. $A+B=\begin{bmatrix}12 & 25\\ 8 & 1\end{bmatrix}$

13. Find k such that $k\begin{pmatrix}2 & 3 & 1\\ 5 & 7 & 4\end{pmatrix}$ is the same as the transpose of $\begin{pmatrix}2 & 5\\ 3 & 7\\ 1 & 4\end{pmatrix}$.
    A. 0
    B. 1
    C. 2
    D. 3

14. Find m and n such that $A=B-C^{T}$,
    where $A=\begin{bmatrix}a & b & c\\ l & m & n\end{bmatrix}$, $B=\begin{bmatrix}6 & 6 & 1\\ 7 & 8 & 8\end{bmatrix}$ and $C=\begin{bmatrix}0 & 5\\ 3 & 4\\ 2 & 4\end{bmatrix}$.
    A. $m=0, n=3$
    B. $m=0, n=0$
    C. $m=3, n=0$
    D. $m=3, n=3$